

$$y = mx + c$$

m：變化率

m: rate of change

c：初始值

c: initial value

x 每增加 1，y 改變多少？

How much does y change for each extra 1 in x?

當  $x = 0$  時，y 是多少？

What is y when  $x = 0$ ?

一條直線描述 x 與 y 的關係

A straight line describes the relationship between x and y

C : 初始值  
c: the initial value

水箱原有 5 公升水

A tank starts with 5 litres of water

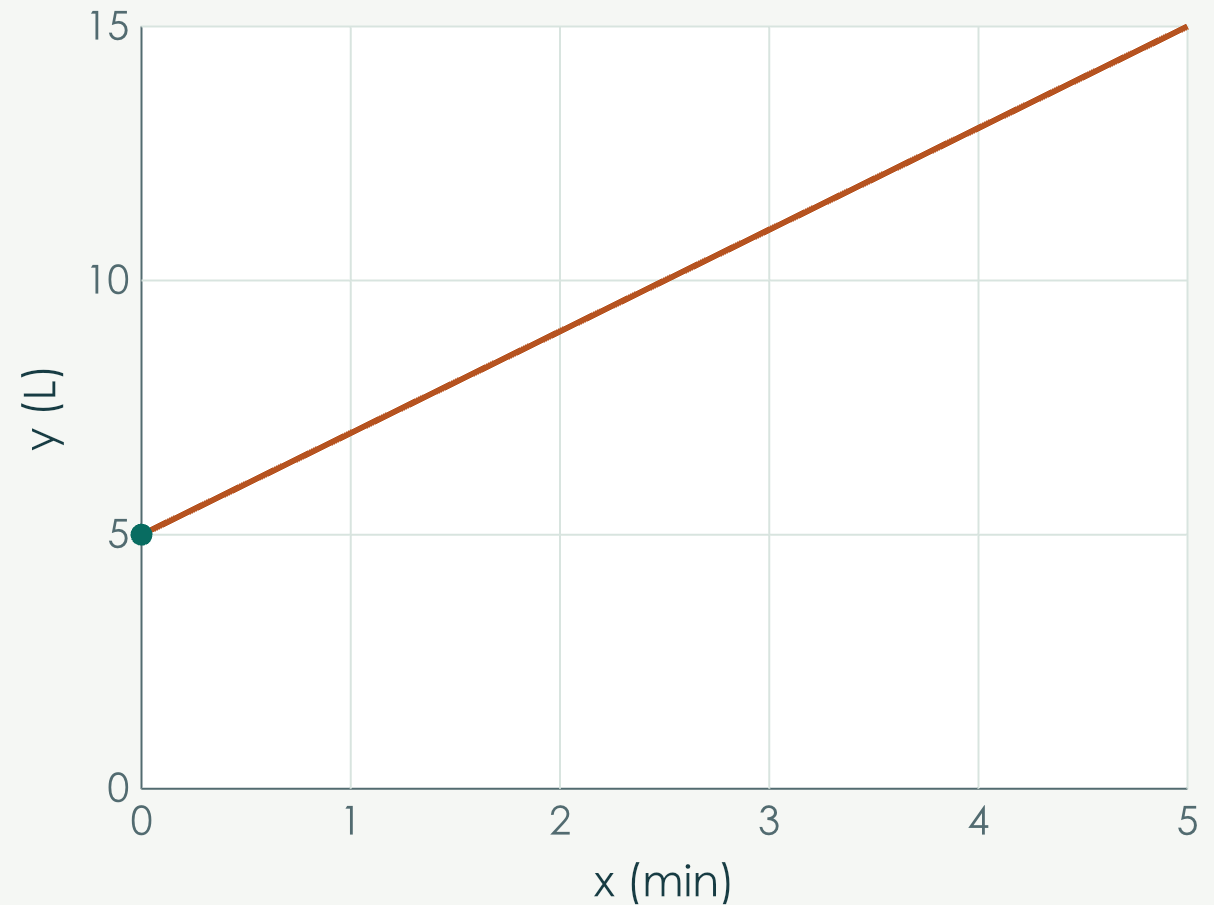
x : 時間 (分鐘)

x: time in minutes

y : 水量 (公升)

y: amount of water in litres

**$x = 0, y = 5$**



m : 每增加 1 單位的變化  
m: the change for each extra unit

每分鐘加入 2 公升水

We add 2 litres each minute

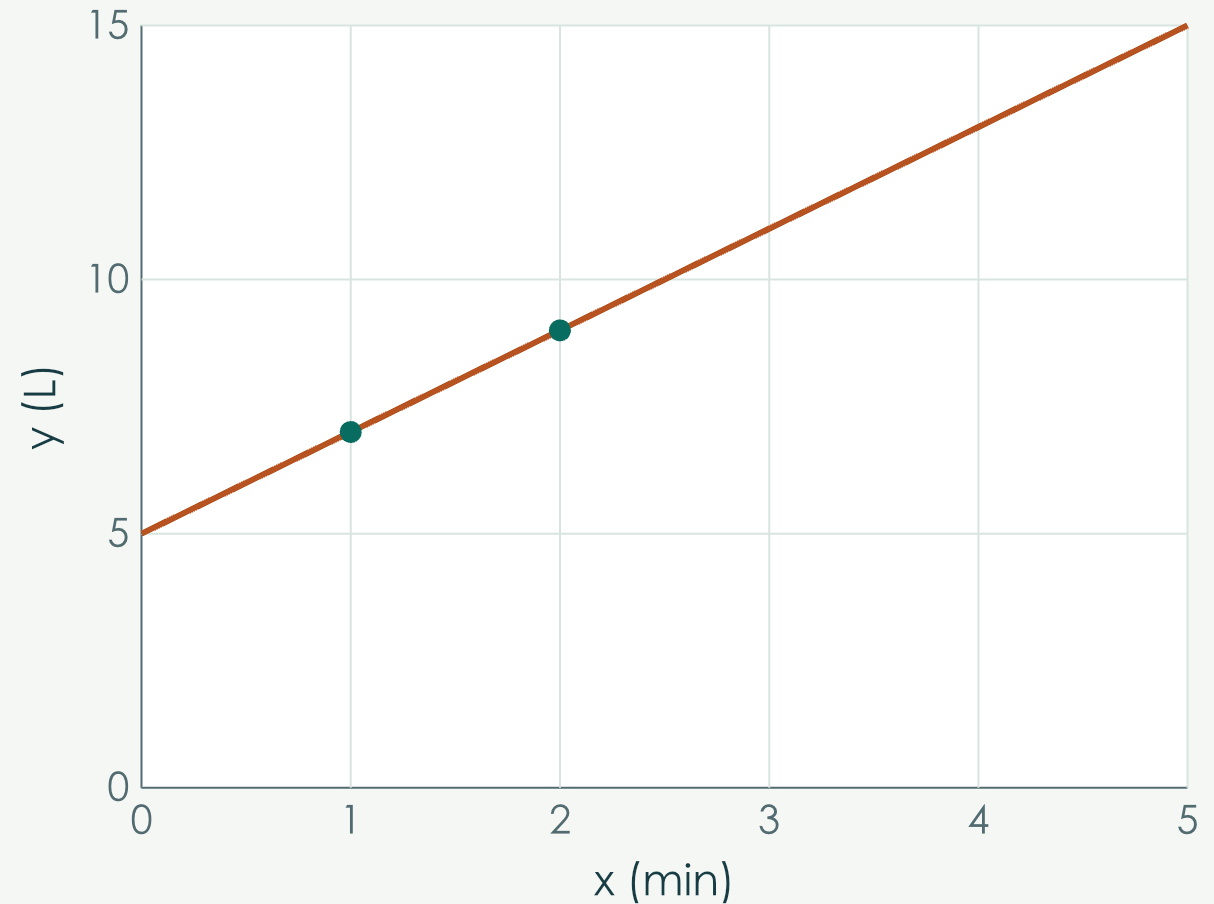
$$y = 2x + 5$$

x 增加 1 , y 增加 2

When x increases by 1, y increases by 2

m = 2 公升 / 分鐘

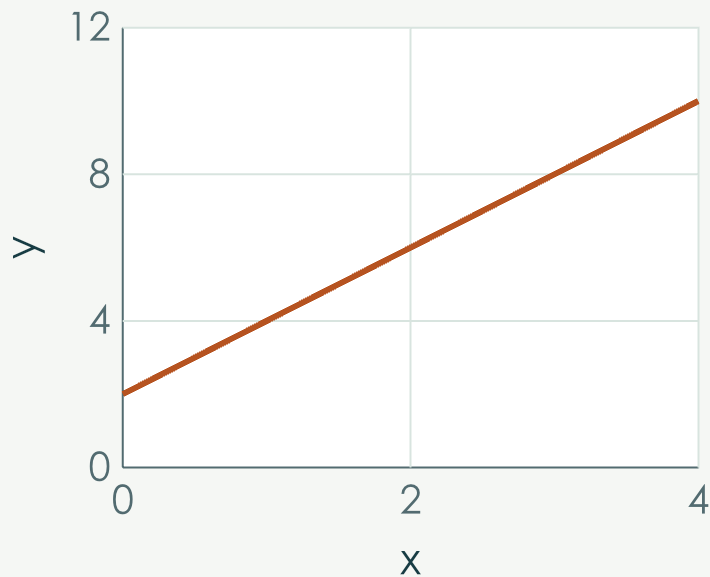
m = 2 litres per minute



# m 的正負與趨勢

The sign of m and the trend

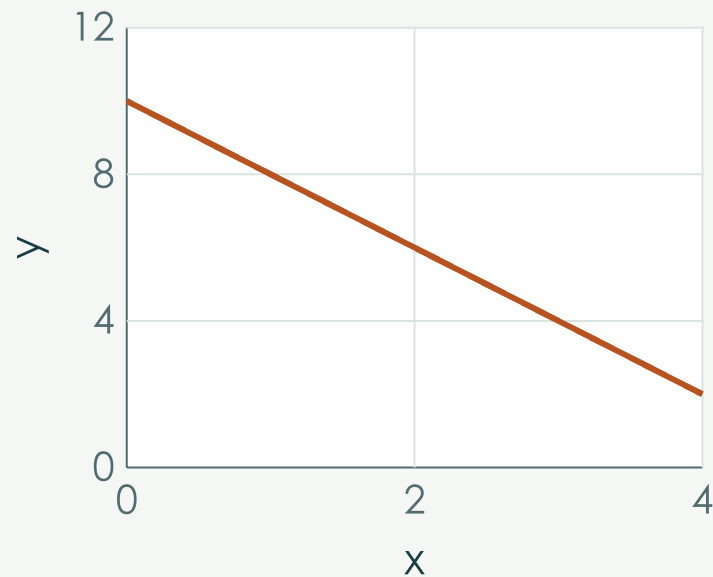
**m > 0**



y 隨 x 增加

y increases as x increases

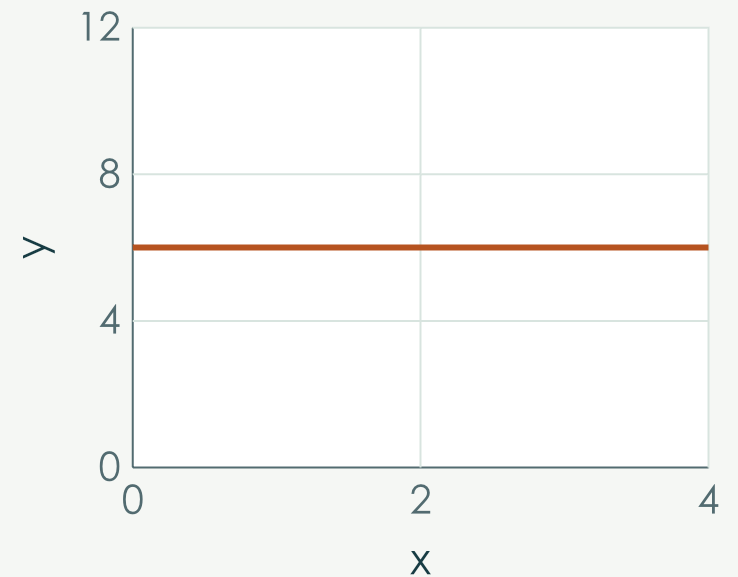
**m < 0**



y 隨 x 減少

y decreases as x increases

**m = 0**

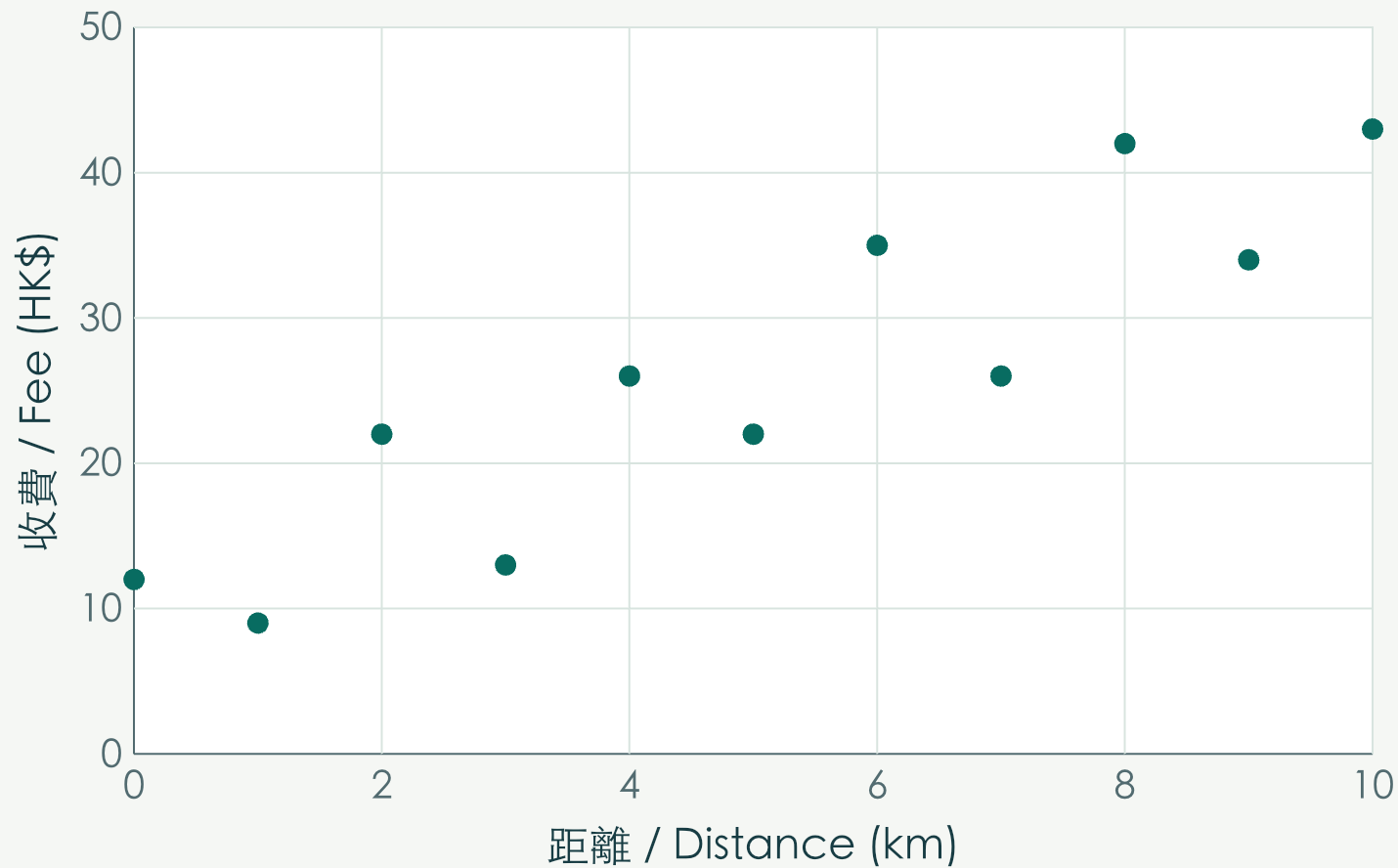


y 保持不變

y stays the same

# 真實情境的數據會有差異

## Data in real situations varies



每個點是一個觀察

Each dot is an observation

各點是否在同一直線上？

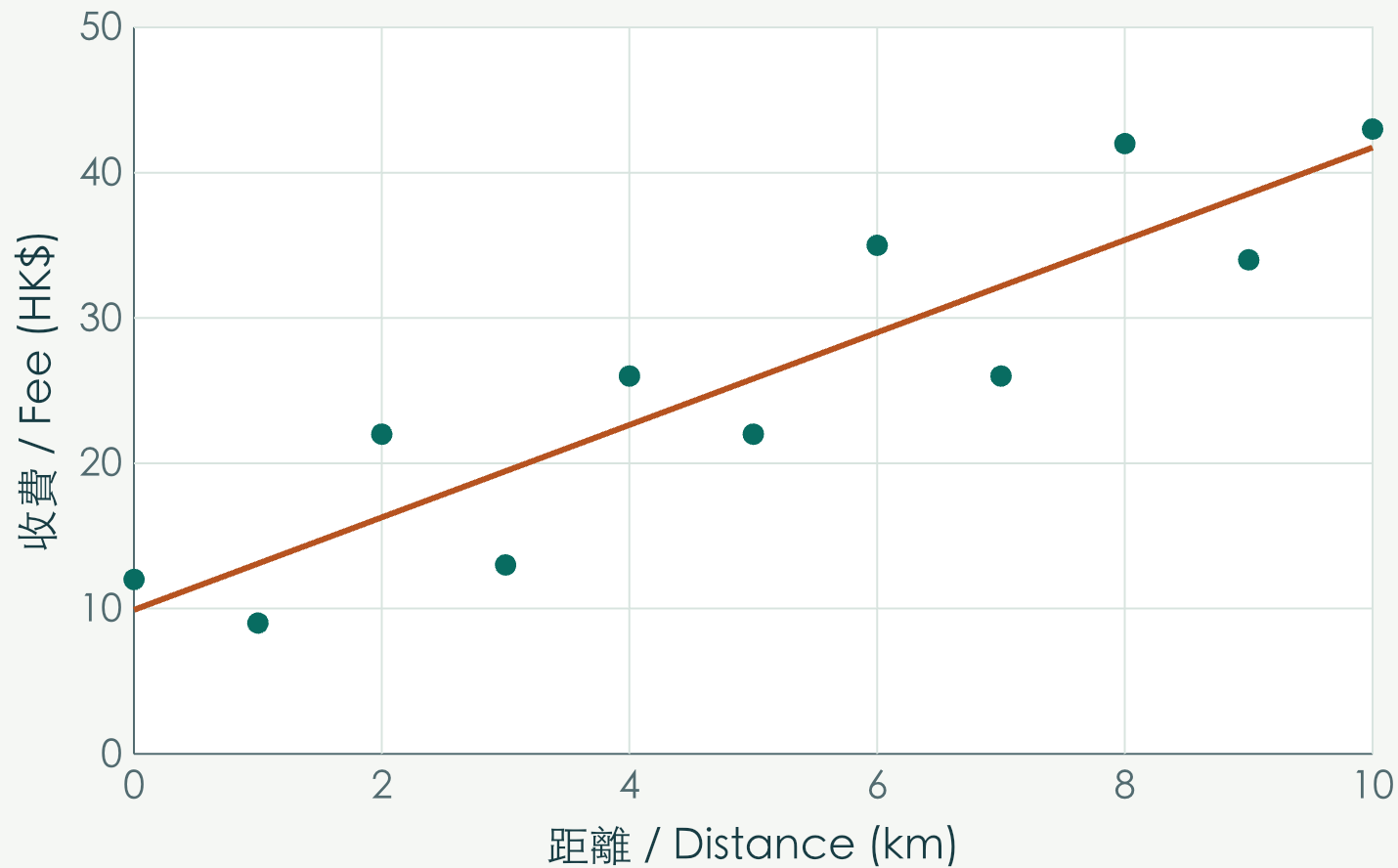
Do all the dots lie on one line?

你看到甚麼趨勢？

What trend do you notice?

# 線性回歸與最佳擬合直線

## Linear regression and the best-fit line



用一條直線概括整體趨勢

One line summarises the overall trend

它不必通過每一個點

It need not pass through every dot

網頁會自動更新直線

The page updates the line automatically

加入新點，觀察直線  
Add points and observe the line

在圖上點一下，加入一個數據點  
Click or tap the graph to add a data point

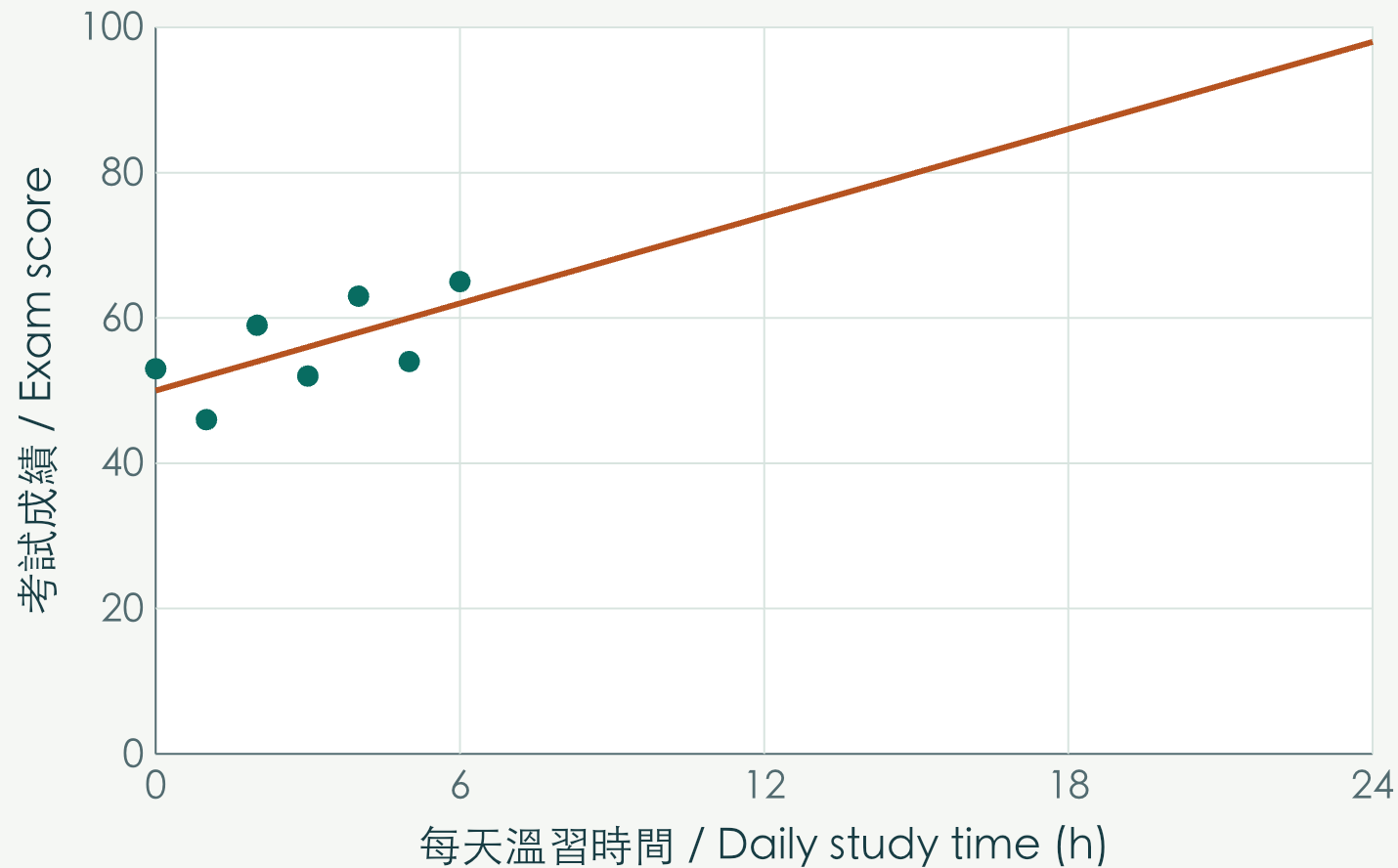
比較加入點前後的  $m$  和  $c$   
Compare  $m$  and  $c$  before and after adding the point

試一個遠離其他點的新點  
Try a point far from the other dots

先預測，再觀察。把結果寫在工作紙上。  
Predict first, then observe. Record the result on your worksheet.

# 每天溫習時間與考試成績

## Daily study time and exam score



3.5 小時的預測成績？

Score for 3.5 hours a day?

先估計，再檢查網頁

Estimate first, then check the page

10 小時的預測可信嗎？

How reliable is a prediction for 10 hours?

不同學生的模擬數據 / Simulated data for different students

# 預測結果與模型的限制

## Prediction results and limits of the model

$$y = 2x + 50$$

$m = 2$  : 模型中每小時約多 2 分

$m = 2$ : about 2 more marks per extra hour

$c = 50$  : 0 小時時估計的成績

$c = 50$ : estimated score at 0 hours

3.5 小時 : 約 57 分

3.5 hours: about 57 marks

10 小時 : 約 70 分

10 hours: about 70 marks

10 超出觀察範圍，可能較不可靠。趨勢不保證個別結果。

10 is outside the observed range. The trend does not guarantee an individual result.

# 你的預測與觀察

## Your prediction and observation

你的估計與網頁的預測接近嗎？

Was your estimate close to the page's prediction?

加入新點後， $m$  和  $c$  有甚麼改變？

How did  $m$  and  $c$  change after you added a point?

你會較信賴哪個預測？為甚麼？

Which prediction would you trust more? Why?

工作紙最後一節 / Final section of the worksheet